

JEE MAIN REVISION MASTER: STRAIGHT LINES AND PAIR OF STRAIGHT LINES 2026 1 QUESTIONS

1.67% WEIGHTAGE ↓49.85%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Straight Lines And Pair Of Straight Lines - Core Application	12 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Straight Lines And Pair Of Straight Lines - Core Application (High Yield — 12 PYQs)

Core Principle: Key problem pattern in Kcet Straight Lines And Pair Of Straight Lines.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The line L_2 passing through $(3,-1)$ divides the line segment L_1 joining the points $(-1,2)$ and $(3,6)$ in the ratio $1:3$ intern..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The system of equations $4x+6y=5$ and $8x+12y=10$ has..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A line passes through $(-1,-3)$ and perpendicular to $x+6y=5$. Its x intercept is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The line L_2 passing through $(3, -1)$ divides the line segment L_1 joining the points $(-1, 2)$ and $(3, 6)$ in the ratio $1:3$ internally. The equation of line L_2 is:

- (A) $4x - 3y - 9 = 0$
- (B) $4x - 3y + 9 = 0$
- (C) $4x + 3y - 9 = 0$
- (D) $4x + 3y + 9 = 0$

Correct Answer: None

Solution Analysis: Let the points on line segment L_1 be $A(-1, 2)$ and $B(3, 6)$. The line L_2 divides this segment internally in the ratio $1:3$. Using the section formula, the point of division P is $P\left(\frac{1 \cdot 3 + 3 \cdot (-1)}{1+3}, \frac{1 \cdot 6 + 3 \cdot 2}{1+3}\right) = P\left(\frac{3-3}{4}, \frac{6+6}{4}\right) = P(0, 3)$. So, line L_2 passes through the points $(3, -1)$ and $(0, 3)$. Now find its slope: $m = \frac{3 - (-1)}{0 - 3} = \frac{4}{-3} = -\frac{4}{3}$. Using point-slope form with point $(0, 3)$: $y - 3 = -\frac{4}{3}(x - 0)$ $3y - 9 = -4x$ $4x + 3y - 9 = 0$. So, the equation of line L_2 is $4x + 3y - 9 = 0$. Correct option